Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.001 with the Adam optimizer. All experiments were run on a single GPU with 24 GB of memory.

2 Results

Figure 1 shows the validation accuracy by dataset.

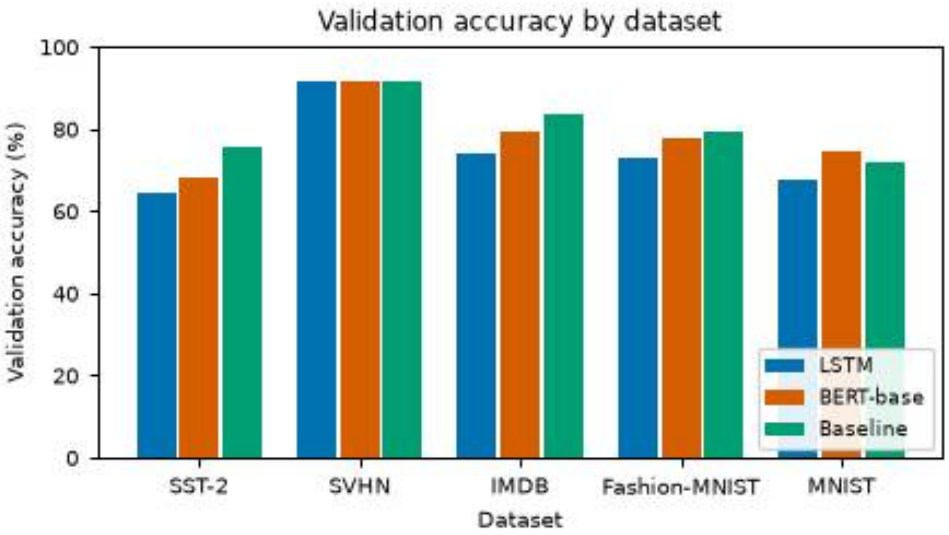


Figure 1: Validation accuracy by dataset for the models in this experiment.

On SST-2, LSTM obtains a validation accuracy of 66.7%. On MNIST, LSTM obtains a validation accuracy of 67.7% (Figure 1). As shown in Figure 1, on SST-2, Baseline is 10.8 points better than LSTM.

Figure 2 shows the validation accuracy by dataset.

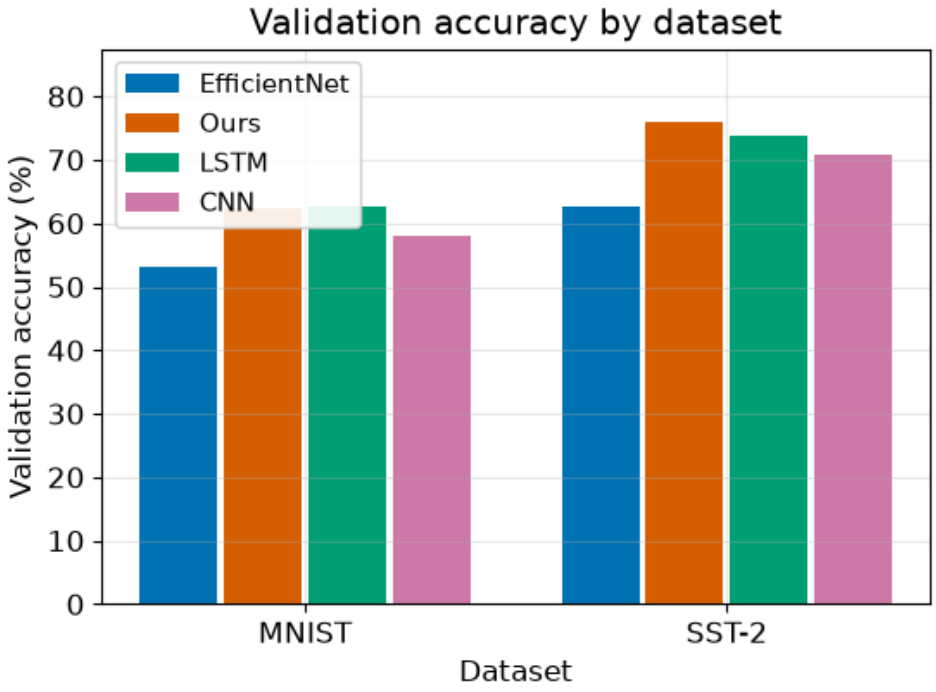


Figure 2: Validation accuracy by dataset for the models in this experiment.

Ours outperforms CNN by 4.4 percentage points on MNIST. CNN reaches 58.0% validation accuracy on MNIST.

Figure 3 shows the accuracy by dataset.

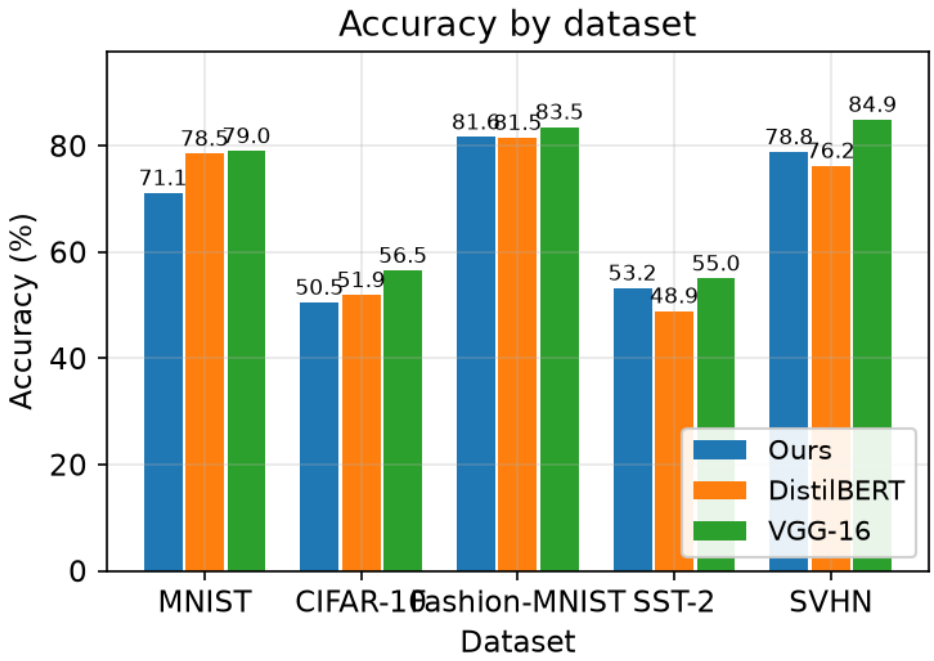


Figure 3: Accuracy by dataset for the models in this experiment.

On Fashion-MNIST, VGG-16 obtains a accuracy of 85.5%. VGG-16 achieves 85.5% accuracy on Fashion-MNIST.

Figure 4 shows the f1 score by dataset.

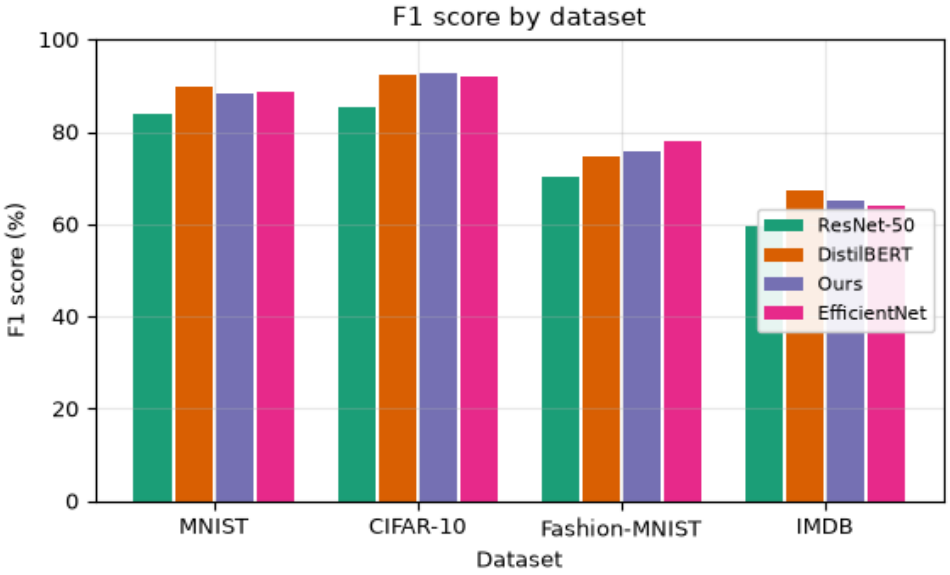


Figure 4: F1 score by dataset for the models in this experiment.

On CIFAR-10, DistilBERT obtains a F1 score of 82.5%. EfficientNet achieves 70.3% F1 score on Fashion-MNIST (Figure 4).

3 Discussion

On MNIST, SVM obtains an accuracy of 86.0%. We note that the training was stable for all models.